

Paul Schmitt

Assistant Professor
Department of Computer Science and Software Engineering
California Polytechnic State University

prs@calpoly.edu
<https://pschmitt.net>

Research Overview

My research focuses on designing and building networked systems that improve security and privacy, enhance performance, and provide insight into networks using emerging machine learning methods. My research is rooted in deep expertise in protocols and architectures that are used at massive scales and underpin modern connectivity. My work achieves broad impact through the design, implementation, deployment, and commercialization of fundamentally new techniques, architectures, systems, and protocols in practice; employing a “dirty-slate” approach to networked systems research for immediate applicability.

Education

Degree	Year	University	Field
Ph.D.	2017	University of California, Santa Barbara Santa Barbara, CA <i>Dissertation:</i> Network Measurement and Systems for Resource-Constrained Environments <i>Advisor:</i> Elizabeth Belding	Computer Science
M.S.	2011	University of St. Thomas St. Paul, MN	Software Engineering
B.A.	2009	University of St. Thomas St. Paul, MN	Computer Science

Positions

Title	Organization	Years
Assistant Professor	California Polytechnic State University	January 2025–Present
Principal Investigator	International Computer Science Institute	June 2024–Present
Assistant Professor	University of Hawai‘i at Mānoa	August 2022–December 2024
Co-Founder and CEO	INVISV	August 2020–Present
Research Computer Scientist	Information Sciences Institute	July 2021–May 2022
Associate Research Scholar	Princeton University	June 2019–July 2021
Postdoctoral Researcher	Princeton University	July 2017–June 2019
Research Assistant (Ph.D.)	University of California, Santa Barbara	2012–2017

Publications

Conference and Journal Publications

- [1] Andrew Chu, Xi Jiang, Shinan Liu, Arjun Bhagoji, Francesco Bronzino, Paul Schmitt, and Nick Feamster. NetSSM: Multi-flow and state-aware network trace generation using state-space models. *Proc. ACM Netw.*, 4(CoNEXT1), March 2026.
- [2] Ayoub Ben Ameer, Francesco Bronzino, Nick Feamster, and Paul Schmitt. Measuring low latency at scale: A field study of L4S in residential broadband. In *Passive and Active Measurement Conference (PAM)*, Virtual, March 2026.

- [3] Xi Jiang, Shinan Liu, Saloua Naama, Francesco Bronzino, Paul Schmitt, and Nick Feamster. JITI: Dynamic model serving for just-in-time traffic inference. *Proc. ACM Netw.*, 3(CoNEXT4), November 2025.
- [4] Taveesh Sharma, Paul Schmitt, Francesco Bronzino, Nick Feamster, and Nicole Marwell. Beyond data points: Regionalizing crowdsourced latency measurements. In *ACM SIGMETRICS*, Stony Brook, NY, USA, June 2025.
- [5] Johann Hugon, Paul Schmitt, and Francesco Bronzino. The cost of packet loss on ML-based traffic analysis. In *2025 IEEE 31st International Symposium on Local and Metropolitan Area Networks (LANMAN)*, Lille, France, July 2025.
- [6] Xi Jiang, Shinan Liu, Aaron Gember-Jacobson, Arjun Nitin Bhagoji, Paul Schmitt, Francesco Bronzino, and Nick Feamster. NetDiffusion: Network data augmentation through protocol-constrained traffic generation. In *ACM SIGMETRICS*, Venice, Italy, June 2024.
- [7] Shinan Liu, Francesco Bronzino, Paul Schmitt, Arjun Nitin Bhagoji, Nick Feamster, Hector Garcia Crespo, Timothy Coyle, and Brian Ward. LEAF: navigating concept drift in cellular networks. *Proc. ACM Netw.*, 1(CoNEXT2), September 2023.
- [8] Martino Trevisan, Idilio Drago, Paul Schmitt, and Francesco Bronzino. Measuring the performance of iCloud Private Relay. In *Passive and Active Measurement Conference (PAM)*, virtual, March 2023.
- [9] Francesco Bronzino, Paul Schmitt, Sara Ayoubi, Hyojoon Kim, Renata Teixeira, and Nick Feamster. Traffic refinery: Cost-aware traffic representation for machine learning in networks. In *ACM SIGMETRICS*, June 2022.
- [10] Jordan Holland, Paul Schmitt, Nick Feamster, and Prateek Mittal. New directions in automated traffic analysis. In *ACM Conference on Computer and Communications Security (CCS)*, November 2021.
- [11] Paul Schmitt and Barath Raghavan. Pretty good phone privacy. In *USENIX Security Symposium*, August 2021.
- [12] Austin Hounsel, Paul Schmitt, Kevin Borgolte, and Nick Feamster. Can encrypted DNS be fast? In *Passive and Active Measurement Conference (PAM)*, Brandenburg, Germany, March 2021.
- [13] Shinan Liu, Paul Schmitt, Francesco Bronzino, and Nick Feamster. Characterizing service provider response to the COVID-19 pandemic in the United States. In *Passive and Active Measurement Conference (PAM)*, Brandenburg, Germany, March 2021.
- [14] Francesco Branzino, Nick Feamster, Shinan Liu, James Saxon, and Paul Schmitt. Mapping the Digital Divide: Before, During, and After COVID-19. In *Research Conference on Communications, Information and Internet Policy (TPRC)*, Washington, DC, February 2021.
- [15] Francesco Bronzino, Paul Schmitt, Sara Ayoubi, Guilherme Martins, Renata Teixeira, and Nick Feamster. Inferring Streaming Video Quality from Encrypted Traffic: Practical Models and Deployment Experience. In *ACM SIGMETRICS*, Boston, Massachusetts, USA, June 2020.
- [16] Austin Hounsel, Kevin Borgolte, Paul Schmitt, Jordan Holland, and Nick Feamster. Comparing the Effects of DNS, DoT, and DoH on Web Performance. In *The Web Conference (WWW)*, Taipei, Taiwan, April 2020.
- [17] Kevin Borgolte, Tithi Chattopadhyay, Nick Feamster, Mihir Kshirsagar, Jordan Holland, Austin Hounsel, and Paul Schmitt. How DNS over HTTPS is Reshaping Privacy, Performance, and Policy in the Internet Ecosystem. In *Research Conference on Communications, Information and Internet Policy (TPRC)*, Washington, DC, September 2019.
- [18] Paul Schmitt, Anne Edmundson, and Nick Feamster. Oblivious DNS: Practical Privacy for DNS Queries. In *Symposium on Privacy Enhancing Technologies (PETS)*, Stockholm, Sweden, July 2019.
- [19] Paul Schmitt, Daniel Iland, Mariya Zheleva, and Elizabeth Belding. Third-party cellular congestion detection and augmentation. *IEEE Transactions on Mobile Computing*, 18(1), 2019.

- [20] Vivek Adarsh, Paul Schmitt, and Elizabeth Belding. MPTCP performance over heterogenous sub-paths. In *International Conference on Computer Communication and Networks (ICCCN)*, Valencia, Spain, July 2019.
- [21] M. Zheleva, T. Larock, P. Schmitt, and P. Bogdanov. Airpress: High-accuracy spectrum summarization using compressed scans. In *IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, Santa Clara, California, USA, October 2018.
- [22] Paul Schmitt, Francesco Branzino, Renata Teixeira, Tithi Chattopadhyay, and Nick Feamster. Enhancing Transparency: Internet Video Quality Inference from Network Traffic. In *Research Conference on Communications, Information and Internet Policy (TPRC)*, Washington, DC, September 2018.
- [23] Carleen Maitland, Richard Caneba, Paul Schmitt, and Tom Koutsy. A Cellular Network Radio Access Performance Measurement System: Results from a Ugandan Refugee Settlements Field Trial. In *Research Conference on Communications, Information and Internet Policy (TPRC)*, Washington, DC, September 2018.
- [24] Mariya Zheleva, Petko Bogdanov, Timothy Larock, and Paul Schmitt. AirVIEW: Unsupervised transmitter detection for next generation spectrum sensing. In *IEEE INFOCOM*, Honolulu, Hawaii, USA, April 2018.
- [25] Paul Schmitt, Daniel Iland, and Elizabeth Belding. Smartcell: Small-scale mobile congestion awareness. *Communications Magazine*, 54(7):4450, July 2016.
- [26] Paul Schmitt, Daniel Iland, Elizabeth Belding, and Mariya Zheleva. Phonehome: Robust extension of cellular coverage. In *International Conference on Computer Communication and Networks (ICCCN)*, Waikoloa, Hawaii, USA, August 2016.
- [27] Paul Schmitt, Daniel Iland, Elizabeth Belding, Brian Tomaszewski, Ying Xu, and Carleen Maitland. Community-level access divides: A refugee camp case study. In *International Conference on Information and Communications Technologies and Development (ICTD)*, Ann Arbor, Michigan, USA, June 2016.
- [28] Paul Schmitt, Daniel Iland, Mariya Zheleva, and Elizabeth Belding. Hybridcell: Cellular connectivity on the fringes with demand-driven local cells. In *IEEE INFOCOM*, San Francisco, California, USA, April 2016.
- [29] Paul Schmitt, Morgan Vigil, and Elizabeth Belding. A study of MVNO data paths and performance. In *Passive and Active Measurement Conference (PAM)*, Heraklion, Crete, Greece, March 2016.
- [30] Mariya Zheleva, Paul Schmitt, Morgan Vigil, and Elizabeth Belding. Internet bandwidth upgrade: Implications on performance and usage in rural Zambia. *Information Technologies & International Development*, 11(2), 2015.
- [31] Paul Schmitt, Ramya Raghavendra, and Elizabeth Belding. Internet media upload caching for poorly-connected regions. In *ACM Symposium on Computing for Development (DEV)*, London, United Kingdom, December 2015.
- [32] Mariya Zheleva, Paul Schmitt, Morgan Vigil, and Elizabeth Belding. The increased bandwidth fallacy: Performance and usage in rural Zambia. In *ACM Symposium on Computing for Development (DEV)*, Cape Town, South Africa, December 2013.

Workshop Publications

- [33] Augustin Laouar, Loïc Desgeorges, Paul Schmitt, and Francesco Branzino. Rethinking geolocalization on the internet. In *ACM SIGCOMM Workshop on Hot Topics in Networking (HotNets)*, November 2025.
- [34] Andrew Chu, Xi Jiang, Shinan Liu, Arjun Nitin Bhagoji, Francesco Branzino, Paul Schmitt, and Nick Feamster. Feasibility of state space models for network traffic generation. In *ACM SIGCOMM Workshop on Networks for AI Computing*, August 2024.

- [35] Xi Jiang, Shinan Liu, Aaron Gember-Jacobson, Paul Schmitt, Francesco Bronzino, and Nick Feamster. Generative, high-fidelity network traces. In *ACM SIGCOMM Workshop on Hot Topics in Networking (HotNets)*, November 2023.
- [36] Paul Schmitt, Jana Iyengar, Christopher Wood, and Barath Raghavan. The decoupling principle: A practical privacy framework. In *ACM SIGCOMM Workshop on Hot Topics in Networking (HotNets)*, November 2022.
- [37] Xi Jiang, Shinan Liu, Saloua Naama, Francesco Bronzino, Paul Schmitt, and Nick Feamster. Towards designing robust and efficient classifiers for encrypted traffic in the modern Internet. In *Internet Architecture Board (IAB) Management Techniques in Encrypted Networks Workshop*, 2022.
- [38] Austin Hounsel, Paul Schmitt, Kevin Borgolte, and Nick Feamster. Designing for tussle in encrypted DNS. In *ACM SIGCOMM Workshop on Hot Topics in Networking (HotNets)*, November 2021.
- [39] Austin Hounsel, Paul Schmitt, Kevin Borgolte, and Nick Feamster. Encryption without Centralization: Distributing DNS Queries Across Recursive Resolvers. In *ACM/IRTF Applied Networking Research Workshop (ANRW)*, July 2021.
- [40] Francesco Bronzino, Elizabeth Culley, Nick Feamster, Shinan Liu, Jason Livingood, and Paul Schmitt. Interconnection changes in the United States. In *Internet Architecture Board (IAB) COVID-19 Workshop*, 2020.
- [41] Paul Schmitt, Anne Edmundson, Allison Mankin, and Nick Feamster. Oblivious DNS: Practical Privacy for DNS Queries. In *IRTF Applied Networking Research Workshop (ANRW)*, Montreal, Quebec, Canada, July 2019.
- [42] Austin Hounsel, Kevin Borgolte, Paul Schmitt, Jordan Holland, and Nick Feamster. Analyzing the Costs (and Benefits) of DNS, DoT, and DoH for the Modern Web. In *IRTF Applied Networking Research Workshop (ANRW)*, Montreal, Quebec, Canada, July 2019.
- [43] Paul Schmitt and Elizabeth Belding. Low on air: Inherent wireless channel capacity limitations. In *ACM Workshop on Computing Within Limits (LIMITS)*, Santa Barbara, California, USA, June 2017.
- [44] Thomas Pötsch, Paul Schmitt, Jay Chen, and Barath Raghavan. Helping the lone operator in the vast frontier. In *ACM SIGCOMM Workshop on Hot Topics in Networking (HotNets)*, Atlanta, Georgia, USA, November 2016.
- [45] Paul Schmitt and Elizabeth Belding. Navigating connectivity in reduced infrastructure environments. In *ACM Workshop on Computing Within Limits (LIMITS)*, Irvine, California, USA, June 2016.
- [46] Mariya Zheleva, Paul Schmitt, Morgan Vigil, and Elizabeth Belding. Community detection in cellular network traces. In *International Conference on Information and Communications Technologies and Development (ICTD: Notes)*, Cape Town, South Africa, December 2013.
- [47] Mariya Zheleva, Paul Schmitt, Morgan Vigil, and Elizabeth Belding. Bringing visibility to rural users in Cote d'Ivoire. In *International Conference on Information and Communications Technologies and Development (ICTD: Notes)*, Cape Town, South Africa, December 2013.
- [48] Paul Schmitt, Morgan Vigil, Mariya Zheleva, and Elizabeth Belding. Communication flow patterns in the D4D dataset. In *NetMob Session on Data for Development (D4D)*, Boston, Massachusetts, USA, April 2013.

Books and Book Chapters

- [49] Paul Schmitt, Daniel Iland, Elizabeth Belding, and Mariya Zheleva. *Digital Lifeline? ICTs for Refugees and Displaced Persons*. MIT Press, 2018. *Chapter: Cellular and Internet Connectivity for Displaced Populations*.

Thesis

- [50] Paul Schmitt. *Network Measurement and Systems for Resource-Constrained Environments*. PhD thesis, University of California, Santa Barbara, June 2017.

Technical Reports

- [51] Sulagna Mukherjee, Srivatsan Ravi, Paul Schmitt, and Barath Raghavan. The ghost trilemma.
- [52] Jay Chen, Barath Raghavan, Paul Schmitt, and Tai Liu. A pluralist approach to democratizing online discourse arXiv:2108.12573 [cs.NI], 2021.
- [53] Jordan Holland, Ross Teixeira, Paul Schmitt, Kevin Borgolte, Jennifer Rexford, Nick Feamster, and Jonathan Mayer. Classifying network vendors at Internet scale arXiv:2006.13086 [cs.NI], 2020.
- [54] Bilal Saleem, Paul Schmitt, Jay Chen, and Barath Raghavan. Beyond the trees: Resilient multipath for last-mile WISP networks. arXiv:2002.12473 [cs.NI], 2020.
- [55] Anne Edmundson, Paul Schmitt, Nick Feamster, and Jennifer Rexford. OCDN: Oblivious content distribution networks. arXiv:1711.01478 [cs.NI], 2017.
- [56] Adam Lerner, Giulia Fanti, Yahel Ben-David, Jesus Garcia, Paul Schmitt, and Barath Raghavan. Rangzen: Anonymously getting the word out in a blackout. arXiv:1612.03371 [cs.NI], 2016.
- [57] Carleen Maitland, Brian Tomaszewski, Elizabeth Belding, Karen Fisher, Ying Xu, Danny Iland, Paul Schmitt, and Amira Majid. Youth mobile phone and Internet use, January 2015, Zaatari Camp, Mafraq, Jordan. Technical report, Penn State University, State College, Pennsylvania, USA, 2015.

Patents

- [Secure network routing as a service](#)
Paul Schmitt, Barath Raghavan
United States Patent No. US20230344808A1
- [Wireless device privacy within wireless mobile](#)
Barath Raghavan, Paul Schmitt
United States Patent No. US20230292127A1
- [System and method for phone privacy](#)
Paul Schmitt, Barath Raghavan
PCT Patent No. WO2022060498A1
- [Local content sharing through edge caching and time-shifted uploads](#)
Kang-Won Lee, Robert Nicholson, Ramya Raghavendra, Paul Schmitt, Dinesh Verma
United State Patent No. 10114828B2

Selected Invited Talks

- “Networked Systems for a Modern, Private Internet.” *Carnegie Mellon University*. Pittsburgh, PA, March 2022.
- “Pretty Good Phone Privacy.” *Telefónica Research*. Madrid, Spain, October 2021.
- “Internet Video Quality Inference from Encrypted Network Traffic.” *NSF Workshop on Measurements for Self-Driving Networks*. Princeton, NJ, April 2019.
- “Internet Video Quality Inference from Encrypted Network Traffic.” *Network Programming Initiative Fall Retreat*. New York, NY, October 2018
- “ODNS: Oblivious DNS.” *DNS-OARC 28*. San Juan, Puerto Rico, March 2018.
- “Correlating Network Congestion with Video QoE Degradation - a Last-Mile Perspective.” *Workshop on Active Internet Measurements (AIMS)*. San Diego, CA, March 2018.

Selected Press

- “New Mobile Phone Service Shows We Can Have Both Privacy and Nice Things” *ACLU*. February 2023.
- “A Phone Carrier That Doesn’t Track Your Browsing or Location” *WIRED*. August 2022.
- “A Simple Software Fix Could Limit Location Data Sharing” *WIRED*. August 2021.
- “The Truth About Faster Internet: It’s Not Worth It” *Wall Street Journal*. Front Page (A1) Feature. August 2019.
- “Sorry spooks: Princeton boffins reckon they can hide DNS queries” *The Register*. April 2018.
- “Isolated in Zaatari camp, Syrian refugees find ways to get online” *Al Jazeera America*. July 2015.

Teaching and Mentoring

- Introduction to Computer Networks (CPE 464), Cal Poly, Fall 2025
- Operating Systems (CSC 453), Cal Poly, Winter, Spring 2025
- Computer Communications Networks (EE 449), UH Mānoa, Spring 2023, Spring 2024
- Privacy and Security in the Modern Internet (EE 693F), UH Mānoa, Fall 2022, Fall 2023, Fall 2024
- Object Oriented Design and Implementation [TA] (CMPSC 32), UCSB, Spring 2013
- Network Computing [TA] (CMPSC 176b), UCSB, Winter 2013
- Object Oriented Design and Implementation [TA] (CMPSC 32), UCSB, Fall 2012
- Computer Networking (CISC 370), University of St. Thomas, Spring 2012
- Operating Systems (CISC 310), University of St. Thomas, Fall 2011

Software

Ongoing and Maintained Software

- *L4Proxy* (<https://github.com/Invisv-Privacy/l4proxy>). L4Proxy is a high-performance first-hop MASQUE proxy Network Function Virtualization (NFV) module. L4Proxy provides server-side functionality needed for running a Multi-Party Relay service to protect users’ network privacy.
- *nPrint* (nprint.github.io/nprint). nPrint is a standard data representation for network traffic meant to be directly usable with machine learning algorithms, replacing feature engineering for a wide array of traffic analysis problems. nPrint encodes data in a normalized, bit-aligned, space-efficient format that facilitates representation learning which automatically discovers the semantically important portions of network packets.
- *NetMicroscope* (netmicroscope.com). NetMicroscope supports passive measurement for a wide variety of services and network scenarios. The system has two components: (1) a packet processing module that measures flow statistics and tracks their state at line rate; and (2) a quality inference module that queries the flow cache to obtain flow statistics and derive the service quality for tracked applications. We deployed this software in 50 user homes in the United States and France and worked with the Wall Street Journal to publish investigations made possible by NetMicroscope, including how video quality relates to access ISP speed. NetMicroscope led to the creation of a startup.

Older Projects

- *PGPP* Pretty Good Phone Privacy (PGPP) is a fundamentally new cellular network architecture that provides users with location anonymity from the cellular carrier itself. By implementing our design, a cellular carrier no longer has the ability to track and record (and subsequently monetize) individual users’ location history. PGPP is a core technology behind *INVISV*, a venture backed privacy-focused startup.

- *Oblivious DNS*. ODNS increases DNS privacy that (1) obfuscates the queries that a recursive resolver sees from the clients that issue DNS queries; and (2) obfuscates the client's IP address from upper levels of the DNS hierarchy that ultimately resolve the query (i.e., the authoritative servers). ODNS operates in the context of the existing DNS protocol, allowing the existing deployed infrastructure to remain unchanged. ODNS resulted in an IETF draft ("Oblivious DNS - Strong Privacy for DNS Queries"), has led to multiple industry-authored IETF drafts ("Oblivious DNS Over HTTPS", "Adaptive DNS Privacy"), and its underlying techniques have been implemented by Cloudflare, Apple, and Fastly, among others¹.
- *HybridCell*. A system that enables coexistence between commercial and local cellular coverage to augment poorly performing infrastructure. HybridCell monitors nearby commercial cellular networks and dynamically adjusts local network usage based on the "health" of the commercial network.
- *Cellular congestion detection*. A system and algorithm for passive detection and characterization of cellular base station radio congestion. The system has been used in measurement campaigns in rural Guatemala and large refugee camps in Uganda and Jordan by the United Nations and USAID. Our findings allowed local organizations to approach carriers to improve connectivity.
- *AirPress*. A system that allows for one-pass spectrum characterization. AirPress employs wavelet decomposition to achieve rapid transmitter detection in low and high noise environments.
- *WISPR*. A network protocol, WISPR (Wireless Internet Service Providers with Redundancy), that enables multi-path over low-cost, heterogeneous wireless links.
- *VillageCache*. A system that transparently scrapes uploads and redelivers locally-created media in users' web sessions without accessing the Internet bottleneck link for poorly-connected networks.

Professional Service

- ACM Workshop on Hot Topics in Networks (HotNets) 2025: TPC
- ACM Internet Measurement Conference (IMC) 2025: TPC
- ACM CoNEXT 2025: Publication Co-Chair
- ACM The Web Conference (WWW) 2024: Reviewer
- Passive and Active Measurement Conference (PAM) 2023: TPC
- The International Teletraffic Congress (ITC) 2023: Reviewer
- NSF review panel 2020
- ACM Journal on Computing and Sustainable Societies: Reviewer
- IEEE INFOCOM 2019: Reviewer
- ACM MobiCom 2017: Reviewer
- IEEE Transactions on Mobile Computing: Reviewer
- IEEE International Conference on Data Mining 2014: Reviewer

Department and University Service

- Dedicated student advisor, Cal Poly Computer Science and Software Engineering, 2025-Present
- Curriculum committee, Cal Poly Computer Science and Software Engineering, 2024-25
- Graduate committee, UH Mānoa Electrical and Computer Engineering, 2022-Present
- Faculty hiring committee, UH Mānoa Electrical and Computer Engineering, 2022-23, 2023-24
- Business office working group, UH Mānoa College of Engineering, 2023-Present

¹<https://blog.cloudflare.com/oblivious-dns/>

Honors and Awards

- 2024 Comcast Innovation Fund Award
- 2016-2017 UCSB President's Dissertation Year Fellowship
- 2014-2015 Dean's Fellowship, UCSB